



Severity of e-scooter rider injuries associated with trip characteristics

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ABSTRACT

Introduction: E-scooter rider injuries have been growing, but little is known about how trip and incident characteristics contribute to their severity. **Method:** We enrolled 105 adults injured while riding e-scooters who presented to an emergency department in Washington, DC, during 2019. Enrolled participants completed an interview during the emergency department visit, and their charts were abstracted to document their injuries and treatment. Logistic regression examined the association of incident location and circumstances with the likelihood of sustaining an injury on the Abbreviated Injury Scale (AIS) ≥ 2 , while controlling for rider characteristics. **Results:** The most common locations of e-scooter injuries in our study sample occurred on the sidewalk (58%) or road (23%). Accounting for other trip and rider attributes, e-scooter riders injured on the road were about twice as likely as those injured elsewhere to sustain AIS ≥ 2 injuries (RR, 1.96; 95% CI, 1.23–2.36) and those who rode at least weekly more often sustained AIS ≥ 2 injuries compared with less frequent riders (RR, 1.86; 95% CI, 1.11–2.32). **Conclusions:** Greater injury severity for riders injured on the road may reflect higher travel speeds. **Practical applications:** Injury severity associated with riding in the road is one factor that jurisdictions can consider when setting policy on where e-scooters should be encouraged to ride, but the risk of any crash or fall associated with facilities should also be examined. Although injuries are of lower severity on sidewalks, sharing sidewalks with slower moving pedestrians could potentially lead to more conflicts.

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1. Introduction

Shared micromobility first appeared in the United States with a fleet of 120 bicycles at 10 docking stations in Washington, DC, in 2008 (Capital Bikeshare, 2020). By 2019, 136 million annual trips were made on shared manual bicycles, e-bikes, and e-scooters, with more than half of these trips made on e-scooters (National Association of City Transportation Officials, 2020). Although the future of e-scooters after the COVID-19 pandemic is uncertain (Hawkins, 2020), they may reemerge in cities as a necessary transportation option if the public is hesitant about resuming transit use (Lazo, 2020).

As the number of e-scooter trips has increased, so has the number of injured e-scooter riders treated in U.S. emergency departments (Namiri et al., 2020). A growing body of evidence has documented the types of injuries sustained by e-scooter users treated in emergency departments and admitted to hospitals (e.g., Badeau et al., 2019; Bauer et al., 2020; Dhillon et al., 2020;

English et al., 2020; Puzio et al., 2020; Trivedi et al., 2019). However, less is known about how the circumstances leading to injuries affect the types of injuries sustained. Policies vary widely among cities on if e-scooters should use roads, sidewalks, bike lanes, or multi-use trails (Chang, Miranda-Moreno, Clewlow, & Sun, 2019; Fang, Agrawal, & Hooper, 2019), and to date there has not been research evidence available to guide these decisions.

The objective of this study was to examine the relationship between where and how e-scooter riders are injured and the severity of their injuries. We interviewed e-scooter riders and other individuals injured by e-scooters (i.e., as pedestrians or cyclists) who sought treatment at a Washington, DC, emergency department during 2019. Shared dockless e-scooters became available in Washington, DC, in March 2018. Five operators deployed 3,000 e-scooters that were used on over 300,000 trips during March 2019, when the study began, and during the course of 2019 the program grew to eight operators and over 4,600 scooters, with riders completing more than 600,000 monthly trips during the peak summer months. E-scooters in the District are prohibited from riding on the sidewalk in the city's Central Business District, without other constraints on where they can ride.

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2. Methods

We prospectively recruited 105 injured adult e-scooter riders and 6 non-riders injured by e-scooters who presented for treatment within a week of their injury at the emergency department of George Washington University Hospital, an urban academic medical center with an annual emergency department census of 75,000 patient visits. Research assistants screened 148 potential participants at the emergency department during March to November 2019, between the hours of 9 a.m. and 10 p.m. One hundred and five patients enrolled in the study. Of the 37 individuals who were not enrolled, seven were missed by research assistants prior to screening, 19 were ineligible (injury did not involve e-scooter [$n = 11$], too injured to communicate with research staff [$n = 3$], previously enrolled for the same trip [$n = 1$], under age 18 years [$n = 1$], ineligible for reason not recorded [$n = 3$]), and 11 were eligible but refused to participate. The protocol was approved by the George Washington University Hospital Institutional Review Board.

Trained research assistants administered structured interviews with participants while they were in the emergency department, or in the hospital if admitted and were unable to be enrolled in the emergency department. They received a \$10 gift card for completing the interview. Participants were asked about their personal characteristics and the circumstances of the incident leading to their injury. They additionally marked where their injury occurred on an electronic map. We characterized the type of location where participants were injured as a road, bike lane, sidewalk, multi-use trail or other off-road location, or alley based on their report and review of the site in Google Street View if available. Sidewalks were defined as paths intended for pedestrian use next to the road; we categorized paths that were not adjacent to the road as multi-use trails/off-road locations. If a rider was injured in an intersection, we coded the location they were traveling on prior to reaching the intersection.

During 2019, shared e-scooters in Washington, DC, ($n = 85$) and the neighboring jurisdiction of Arlington, VA, ($n = 1$) were required to be speed-limited at 10 mph. Other neighboring jurisdictions did not limit shared e-scooter speeds at this time, and some injured riders used personal e-scooters that were not subject to speed restrictions. Those riding personal e-scooters ($n = 15$) and one rider injured on a shared e-scooter in Alexandria, VA were grouped as riding scooters not limited to 10 mph. Alexandria did not govern e-scooter speeds prior to 2020. Arlington reported that some operators did not obey the 10-mph cap prior to August 2019, but the injured Arlington rider used a brand that did limit speeds.

A research assistant coded each injury sustained by participants using the Abbreviated Injury Scale (AIS) (Gennarelli & Wodzin, 2008), and coding was validated by a second researcher. AIS scores range from 1 to 6, with 1 indicating minor injury; 2, moderate injury; 3, serious injury; to 6 indicating a virtually non-survivable injury. We classified lacerations, contusions, and abrasions, which are assigned to the external body region when calculating the Injury Severity Score (ISS) (Gennarelli & Wodzin, 2008), as external injuries. Injuries assigned to other body regions excluded external injuries and were otherwise categorized using the AIS rather than ISS body regions.

We compared proportions of e-scooter riders by various characteristics using the χ^2 test or Fisher's exact test when χ^2 assumptions were violated, or the t test to compare means. Stepwise logistic regression was used to model the likelihood of AIS ≥ 2 injuries associated with trip characteristics. Variables were removed from the model individually until the remaining variables had values of $p < 0.15$. For ease of interpretation, odds ratios were transformed to relative risks because sustaining AIS ≥ 2 injuries was

not rare (Zhang & Yu, 1998). Missing values are excluded from proportions and analyses. Two riders who left the emergency department prior to being seen by a physician are excluded from the analyses of injuries and treatments.

3. Results

3.1. Rider trip and demographic characteristics

Table 1 describes the locations where the sample of 105 e-scooter riders were injured and the circumstances leading to their incidents. Riders were most often on the sidewalk prior to injury (58%), followed by the road (23%), a multi-use trail or other off-road location (10%), bike lane (8%), or alley (2%). A bike lane was present in the rider's direction of travel for two riders who were injured on the road (8% of road injuries) and six injured on the sidewalk (10% of sidewalk injuries) among those whose exact injury location was known. A third (34%, $n = 20$) of riders injured on sidewalks at known locations were in the Central Business District, where sidewalk riding is prohibited.

Motor-vehicle crashes were infrequent (Table 1). Falls due to self-reported adverse surface features (e.g., pothole, uneven pavement), infrastructure (e.g., driveway lip), or other reasons led to about two thirds of incidents. Adverse surface features were implicated in similar proportions of incidents on roads (33%), bike lanes (25%), and sidewalks (25%).

Other characteristics of riders and their trips compared between those injured in the road and other locations, struck by motor-vehicles and injured in other circumstances, and riding e-scooters capped at 10 mph and other scooters are summarized in Table 2. The total sample was about half male with mean [SD] age of 39.5 [15.2] years. Riding frequency varied, with nearly half reporting that they rode at least weekly during the past 30 days, and 37% reporting they were injured on their first ride. Forty percent were tourists or visitors who did not reside in the Washington, DC, area. Few wore helmets (2%) or rode with more than one person on the same scooter (4%). About one in eight (12%) reported recent alcohol use prior to their trip or were suspected to be impaired by the medical provider or research assistant; participants were not tested for alcohol or other drugs and were approached for consent after their impairment had cleared. The trips of 27% of riders occurred while they were traveling to or from another form of transportation (e.g., public transportation). Sixteen percent reported that a mechanical issue with the scooter such as defective brakes ($n = 6$, 6%), a sticky throttle ($n = 3$, 3%), or defective handlebars or steering ($n = 3$, 3%) contributed to the incident.

Riders were more likely to be local residents or to report riding at least weekly if they were injured in the road (local: 92% vs. 51%, $p < 0.001$; riding weekly: 67% vs. 39%, $p = 0.02$) or were involved in a motor-vehicle crash (local: 100% vs. 55%, $p = 0.005$; riding weekly: 90% vs. 40%, $p = 0.005$) (Table 2). Those involved in motor-vehicle crashes were younger than other riders (mean [SD] age 27.4 [5.4] vs. 40.7 [15.3] years, $p = 0.007$) and their crashes more often occurred at intersections (70% vs. 31%, $p = 0.03$). Riders injured on the road were more often male (88% vs. 42%, $p < 0.001$), on a commuting trip (42% vs. 20%, $p < 0.03$), or traveling to or from another form of transportation (48% vs. 21%, $p = 0.01$) than riders injured elsewhere.

3.2. Rider injury and treatment characteristics

Table 3 describes rider injury and treatment characteristics. Most sustained minor injuries, with 38% having at least one AIS ≥ 2 injury, and few (9%) were admitted to the hospital or transferred. Half had injuries to the extremities, with upper extremity injuries

Table 1
Circumstances and locations where e-scooter riders were injured [N (Column %)].

Circumstances	Locations					Total n = 105 ^a
	Sidewalk n = 60	Road n = 24	Trail/off-road n = 10	Bike lane n = 8	Alley n = 2	
Involved vehicle or other road user						
Crash with moving vehicle	5 (8%)	3 (12%)	0	2 (25%)	0	10 (10%)
Avoiding moving vehicle	2 (3%)	2 (8%)	0	0	0	4 (4%)
Crash with parked vehicle	0	0	0	1 (12%)	0	1 (1%)
Crash with pedestrian	1 (2%)	0	0	0	0	1 (1%)
Avoiding pedestrian	7 (12%)	1 (4%)	1 (10%)	1 (12%)	0	11 (10%)
Avoiding cyclist	0	0	1 (10%)	1 (12%)	0	2 (2%)
Crash with other scooter	0	1 (4%)	0	0	0	1 (1%)
Did not involve vehicle or other road user						
Surface features	15 (25%)	8 (33%)	1 (10%)	2 (25%)	0	26 (25%)
Infrastructure	11 (18%)	3 (12%)	2 (20%)	1 (12%)	0	17 (16%)
Other fall	16 (27%)	5 (21%)	5 (50%)	0	2 (100%)	28 (27%)
Other circumstances	3 (5%)	1 (4%)	0	0	0	4 (4%)

^a Incident location was unknown for 1 participant.

Table 2
Personal and trip characteristics by location of incident and circumstances [N (Column %) or Mean $\pm$ SD].

Characteristic	Location ^a			Circumstances			Total (n = 105)
	Road (n = 24)	Other location (n = 80)	p value	Crash with moving vehicle (n = 10)	Other circumstances (n = 95)	p value	
Age (years)	38.2 $\pm$ 18.0	40.1 $\pm$ 14.3	0.60	27.4 $\pm$ 5.4	40.7 $\pm$ 15.3	0.007	39.5 $\pm$ 15.2
Male	21 (88%)	34 (42%)	<0.001	5 (50%)	50 (53%)	>0.99	55 (52%)
White, non-Hispanic ^a	17 (71%)	46 (59%)	0.30	4 (40%)	59 (63%)	0.18	63 (61%)
College degree or higher	20 (83%)	52 (65%)	0.09	7 (70%)	66 (69%)	>0.99	73 (70%)
Local resident ^a	22 (92%)	39 (51%)	<0.001	10 (100%)	51 (55%)	0.005	61 (60%)
Rode at least weekly during past 30 days ^a	16 (67%)	31 (39%)	0.02	9 (90%)	38 (40%)	0.005	47 (45%)
Wore helmet	0	2 (2%)	>0.99	1 (10%)	1 (1%)	0.18	2 (2%)
Riding with >1 person on scooter	0	4 (5%)	0.57	0	4 (4%)	>0.99	4 (4%)
Alcohol use or appeared impaired	2 (8%)	11 (14%)	0.73	1 (10%)	12 (13%)	>0.99	13 (12%)
Riding to/from another form of transportation ^a	11 (48%)	17 (21%)	0.01	2 (22%)	26 (27%)	>0.99	28 (27%)
Weekend	5 (21%)	27 (34%)	0.23	1 (10%)	32 (34%)	0.17	33 (31%)
Night (9:00 p.m.–5:59 a.m.) ^a	5 (21%)	9 (11%)	0.31	2 (20%)	12 (13%)	0.62	14 (13%)
Commuting trip	10 (42%)	16 (20%)	0.03	4 (40%)	22 (23%)	0.26	26 (25%)
Intersection ^a	11 (46%)	25 (31%)	0.19	7 (70%)	29 (31%)	0.03	36 (35%)
Mechanical issue ^a	6 (25%)	10 (13%)	0.20	1 (10%)	15 (16%)	>0.99	16 (16%)
Not speed-limited ^a	6 (25%)	11 (14%)	0.22	3 (30%)	14 (15%)	0.36	17 (17%)
On road ^a				3 (30%)	21 (22%)	0.69	24 (23%)
motor-vehicle crash	3 (13%)	7 (9%)	0.69				10 (10%)

^a Unknown values excluded.

(34%) most common. Although head injuries were uncommon (7%), four riders (4%) experienced head injuries of moderate (AIS = 2) severity, including intracranial hemorrhage ($n = 1$, 1%), skull fracture ($n = 2$, 2%), and/or concussion with loss of consciousness ($n = 4$, 4%).

A larger proportion of riders injured on the road sustained AIS ≥ 2 injuries (58% vs. 32%, $p = 0.02$), experienced upper extremity injuries (62% vs. 24%, $p < 0.001$), and received splinting in the emergency department (75% vs. 38%, $p = 0.002$) compared with riders injured on other facilities. In contrast, riders who were struck by motor-vehicles were less likely to have extremity injuries (10% vs. 55%, $p = 0.008$) and receive splinting (0% vs. 53%, $p < 0.001$) than riders injured in other circumstances. They were less likely to have AIS ≥ 2 injuries, although not significantly so (10% vs. 41%, $p = 0.09$). The likelihood of riders experiencing AIS ≥ 2 injuries did not differ significantly by other personal and trip characteristics in individual comparisons (Table 4).

Variables in Table 2 were entered into a stepwise logistic regression model examining the risk of sustaining an AIS ≥ 2 injury, and the five variables in Table 5 remained. The risk of experiencing AIS ≥ 2 injuries remained higher for riders injured on the

road compared with other locations when accounting for other trip and rider factors (relative risk [RR], 1.96; 95% confidence interval [CI], 1.23–2.36). In this model, greater injury severity was also associated with riding at least weekly (RR, 1.86; 95% CI, 1.11–2.32), increasing age (RR, 1.02; 95% CI, 1.00–1.05; $p = 0.06$), and female sex (RR, 1.72; 95% CI, 0.98–2.24; $p = 0.06$), and was less likely when riders were struck by a motor-vehicle (RR, 0.19; 95% CI, 0.02–1.10; $p = 0.07$).

3.3. Non-riders

Six enrolled participants were non-riders injured by e-scooters. Four involved parked e-scooters, and two involved moving e-scooters. Both interactions with moving e-scooters occurred at intersections. In one, an e-scooter struck a pedestrian crossing from the sidewalk as the scooter approached perpendicularly from the road; in the other, a cyclist fell while avoiding an e-scooter when the cyclist was crossing from a bike lane and an e-scooter turned left in front of him or her. Of the four participants injured by parked scooters, two were pedestrians who tripped over scooters, one was a cyclist who struck a parked scooter, and one was a

Table 3Injury and treatment characteristics by location of incident and circumstances [N (Column %) or Mean $\pm$ SD].

Characteristic	Location ^a			Circumstances			
	Road (n = 24)	Other location (n = 78)	p value	Crash with moving vehicle (n = 10)	Other circumstances (n = 93)	p value	Total (n = 103)
Maximum AIS $\geq$ 2 (moderate injury)	14 (58%)	25 (32%)	0.02	1 (10%)	38 (41%)	0.09	39 (38%)
Maximum AIS $\geq$ 3 (serious injury)	1 (6%)	1 (1%)	0.42	0	2 (2%)	>0.99	2 (2%)
Admitted or transferred	4 (17%)	5 (6%)	0.21	0	9 (10%)	0.59	9 (9%)
Wound care in ED	10 (42%)	26 (33%)	0.46	4 (40%)	32 (34%)	0.74	36 (35%)
Laceration repair in ED	3 (12%)	22 (28%)	0.12	2 (20%)	23 (25%)	>0.99	25 (24%)
Splinting in ED	18 (75%)	30 (38%)	0.002	0	49 (53%)	<0.001	49 (48%)
Fracture or dislocation reduction in ED	5 (21%)	7 (9%)	0.15	0	12 (13%)	0.60	12 (12%)
Orthopedic procedure performed in hospital	3 (12%)	2 (3%)	0.08	0	5 (5%)	>0.99	5 (5%)
Head injury	1 (4%)	6 (8%)	>0.99	2 (20%)	5 (5%)	0.14	7 (7%)
Face injury	2 (8%)	7 (9%)	>0.99	0	9 (10%)	0.59	9 (9%)
Chest, abdominal, or spine injury	1 (4%)	2 (3%)	0.56	1 (10%)	2 (2%)	0.27	3 (3%)
Extremity injury	19 (79%)	32 (41%)	0.001	1 (10%)	51 (55%)	0.008	52 (50%)
Upper extremity injury	15 (62%)	19 (24%)	<0.001	0	35 (38%)	0.01	35 (34%)
Lower extremity injury	5 (21%)	13 (17%)	0.76	1 (10%)	17 (18%)	>0.99	18 (17%)
External injury	9 (38%)	45 (58%)	0.08	7 (70%)	48 (52%)	0.33	55 (53%)

Abbreviations: AIS, Abbreviated Injury Score; ED, emergency department.

^a Unknown values excluded.

cyclist who fell while avoiding a parked scooter. Three reported the scooter was upright at the time and one that it was lying down, and three occurred on the sidewalk and one on a multi-use trail.

The mean [SD] age of non-riders was 51.0 [17.8], and four were female. Four experienced extremity injuries, three of which were AIS = 2. Two of these participants were admitted to the hospital and underwent orthopedic procedures. The remaining two non-riders experienced external injuries only.

4. Discussion

We found that e-scooter riders injured on the road were more likely to sustain AIS $\geq$ 2 injuries than those injured in other locations in Washington, DC. Motor-vehicle crashes are strongly associated with higher injury severity among cyclists (Chong, Poulos, Olivier, Watson, & Grzebieta, 2010; Crompton et al., 2015; Heesch, Garrard, & Sahlqvist, 2011; Rivara, Thompson, & Thompson, 1997), and it might be expected that injuries sustained on the road would be of greater severity because they more often involve motor-vehicles. In our sample, this was not the case. Motor-vehicle crashes with e-scooters were uncommon, which is consistent with the low proportion of motor-vehicle involvement that has been reported for injured e-scooter riders in other cities (Bekhit, Le Fevre, & Bergin, 2020; Blomberg, Rosenkrantz, Lippert, & Collatz Christensen, 2019; English et al., 2020; Trivedi et al., 2019), and when they did occur, injuries were minor. However, most fatally injured e-scooter riders worldwide are struck by motor-vehicles (Collaborative Sciences Center for Road Safety, 2020). Countermeasures against motor-vehicle crashes would not prevent many of the injuries treated in the emergency department but could prevent fatalities. Most motor-vehicle crashes in this study occurred at intersections, and treatments such as bike boxes or two-stage left-turn queue boxes could potentially make e-scooters more visible to drivers at intersections (Dill, Monsere, & McNeil, 2012; National Association of City Transportation Officials, 2014), if they are used by e-scooter riders.

One possible explanation for the increased severity for those injured on the road could be that these riders were traveling at higher speeds than those injured elsewhere. Arellano and Fang (2019) recorded speeds of e-scooters traveling on streets, sidewalks, and multi-use paths in San Jose, CA, and found that speeds

were fastest on roads, although it is unclear if this difference would transfer to an environment where most e-scooters are limited to top speeds of 10 mph. Circumstances leading to injuries did not differ appreciably between riders injured in the road and elsewhere, but a larger proportion sustained injuries to the upper extremities on the road, which suggests that injury mechanisms differed (e.g., more falls onto an outstretched hand). Riders injured in the road were more likely male, experienced riders that were commuting. Higher injury severity persisted when accounting for rider characteristics, but other behavioral differences beyond speed may have contributed.

Frequent e-scooter riders also experienced greater injury severity. Cycling frequency is associated with injury severity (Crompton et al., 2015; Heesch et al., 2011), and frequent cyclists report greater confidence and riding at higher speeds than less frequent riders (Dill & McNeil, 2013; Poulos et al., 2015). Similarly to riders injured in the road, frequent e-scooter riders may take more risks that could increase severity when a crash or fall occurs if they share these traits. Age and sex were associated with sustaining AIS $\geq$ 2 injuries but these associations were not statistically significant, possibly due to our limited sample size. These characteristics are related to injury severity in other transportation modes (Crompton et al., 2015; Kim, Ulfarsson, Shankar, & Kim, 2008; Li, Braver, & Chen, 2003; Moore, Schneider, Savolainen, & Farzaneh, 2011) and should be evaluated further in a larger study.

Over a third of our sample were injured on their first ride, which is comparable to the proportion reported among injured e-scooter users in Austin, TX (Austin Public Health, 2019). Although the injuries that inexperienced riders sustained were more frequently minor, it appears likely that riders were injured more often during first trips than on subsequent trips. Injured riders have reported receiving training through the e-scooter operator's phone app (Austin Public Health, 2019), but an app may not be an adequate substitute for hands-on experience. As the public gains experience using e-scooters and there are fewer first-time riders, it is possible that the injury rate per unit of riding exposure will decrease while the severity of injuries sustained will rise.

A strength of this study is its combined use of interview and injury data, but this also meant that a small number of injured riders were excluded because they were too injured to be interviewed in the emergency department, or they presented overnight when research assistants were not present. Although we followed up

Table 4
Maximum AIS ≥ 2 by personal and trip characteristics [N (Row %) or Mean $\pm$ SD].

Characteristic	AIS 0 or 1	AIS ≥ 2	p value
Age (n = 103)	37.9 $\pm$ 14.7	43.1 $\pm$ 15.2	0.09
Sex			0.82
Male (n = 54)	33 (61%)	21 (39%)	
Female (n = 49)	31 (63%)	18 (37%)	
Race/ethnicity ^a			0.78
White, non-Hispanic (n = 62)	38 (61%)	24 (39%)	
Other (n = 39)	25 (64%)	14 (36%)	
College degree or higher			0.10
Yes (n = 72)	41 (57%)	31 (43%)	
No (n = 31)	23 (74%)	8 (26%)	
Local resident ^a			0.28
Yes (n = 60)	34 (57%)	26 (43%)	
No (n = 40)	27 (68%)	13 (32%)	
Riding frequency during past 30 days ^a			0.22
At least weekly (n = 47)	26 (55%)	21 (45%)	
Less often (n = 55)	37 (67%)	18 (33%)	
Wore helmet			0.52
Yes (n = 2)	2 (100%)	0	
No (n = 101)	62 (61%)	39 (39%)	
Riding with > 1 person on scooter			0.29
Yes (n = 4)	4 (100%)	0	
No (n = 99)	60 (61%)	39 (39%)	
Alcohol use or appeared impaired			>0.99
Yes (n = 13)	8 (62%)	5 (38%)	
No (n = 90)	56 (62%)	34 (38%)	
Riding to/from another form of transportation ^a			0.75
Yes (n = 27)	16 (59%)	11 (41%)	
No (n = 75)	47 (63%)	28 (37%)	
Day of week			0.83
Weekend (n = 33)	20 (61%)	13 (39%)	
Weekday (n = 70)	44 (63%)	26 (37%)	
Time of day ^a			0.47
Night (9:00 p.m.–5:59 a.m.) (n = 14)	10 (71%)	4 (29%)	
Other (n = 88)	54 (61%)	34 (39%)	
Trip purpose			0.09
Commuting (n = 25)	12 (48%)	13 (52%)	
Other (n = 78)	52 (67%)	26 (33%)	
Intersection ^a			0.55
Yes (n = 35)	23 (66%)	12 (34%)	
No (n = 67)	40 (60%)	27 (40%)	
Mechanical issue ^a			0.28
Yes (n = 16)	12 (75)	4 (25%)	
No (n = 84)	51 (61%)	33 (39%)	
Scooter speed-limited at 10 mph ^a			0.24
Yes (n = 84)	55 (65%)	29 (35%)	
No (n = 16)	8 (50%)	8 (50%)	

Abbreviation: AIS, Abbreviated Injury Score.

^a Unknown values excluded.**Table 5**
Logistic regression model of associations between trip and rider characteristics and risk of sustaining AIS ≥ 2 injury.

Characteristic	RR (95% CI)
Age (1-year increase)	1.02 (1.00, 1.05)
Sex (female vs. male)	1.72 (0.98, 2.24)
Location (road vs. other)	1.96 (1.23, 2.36)*
Circumstances (struck by motor-vehicle vs. other)	0.19 (0.02, 1.10)
Riding frequency (at least weekly vs. less often)	1.86 (1.11, 2.32)*

Abbreviations: AIS, Abbreviated Injury Score; CI, confidence interval; RR, relative risk.

* $p < 0.05$.

with admitted patients in the hospital to minimize potential data loss and only excluded three who were too injured to communicate, this could have biased results. Few injuries were greater severity than AIS = 2, and patterns may be different in a more seriously injured sample. The small sample size, which restricted statistical power and reduced the precision of the results we report, is another limitation. Data were collected during a timeframe set at the beginning of the study, and due to the changing size of the shared e-scooter fleet it was unclear at the outset how many injured e-scooter riders would be evaluated in that time.

Head injuries were less frequent in our sample than what has been seen in other emergency department studies of e-scooter rider injuries, which have reported head injuries in up to 40% of patients (Badeau et al., 2019; Bekhit et al., 2020; Blomberg et al., 2019; English et al., 2020; Trivedi et al., 2019), although the proportion of riders in our study with major head injuries was within the range of prior studies. There are several possibilities for why this may be the case. It may reflect differences in how head injuries were defined, as we categorized lacerations, contusions, and abrasions as external injuries independent of their location on the body. Differences in recruitment methods between our studies and others, which have evaluated a census of injured e-scooter riders rather than limiting to those who could be interviewed, may have also contributed. Washington, DC's 10 mph top speed for shared e-scooters is uncommon among cities with shared dockless programs, where e-scooters can typically travel up to 15 or 20 mph. The lower speeds could potentially affect risk of head injury and decrease injury severity overall by lowering the impact energy in a crash or fall.

New riders or those that used scooters infrequently were not often injured in the road, which indicates that, similar to cyclists (Stinson & Bhat, 2005), inexperienced e-scooter riders were not comfortable there. This combined with greater injury severity for those riding in the road suggests a need for off-road facilities where e-scooters can ride. It is not clear, though, what types of separated facilities are most beneficial to e-scooter riders and the non-riders that may share them. Injury severity was low on sidewalks and we did not see much evidence of e-scooters injuring pedestrians there in this study, but sharing sidewalks with slower moving pedestrians may lead to a trade-off between reduced severity at lower speeds and more opportunities for conflicts.

Riders report preferring using bike lanes to sidewalks (City of Austin, 2019; Portland Bureau of Transportation, 2018); however, far fewer riders were injured on bike lanes than sidewalks in this sample. This could be because e-scooters did not use bike lanes due to unavailability in the parts of the city where tourists rode on the sidewalk (e.g., the National Mall) or discomfort with the speed differential with bicycles. It is also possible that the risk of crashing or falling was lower in bike lanes. To develop robust policy recommendations on where e-scooters should ride, the risk of sustaining an injury needs to be established and considered alongside the severity of injuries experienced.

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